FORKLIFT

PORTALDOT HACKER HOUSE 2026 · NODE OPERATIONS & TOOLING

# YOUR TRANSACTION RUNS TWICE. THE FIRST TIME IS A REHEARSAL.




| KEYS LIFTED · #118,061 | SAMPLED KEYS IDENTICAL | V3 SENDS, SAME TX HASH |
|---|---|---|
| 5,508 | 200/200 | 3 |

```
$ forklift rehearse Balances.transfer_keep_alive … --send
chain     #118,621 118621-1 ok · finalized · tx 0x2247…c36f
compare   8/14 fields match · DIFFERS: …,
          inclusion_height, fee, …
$ forklift replay 118621-1
compare   14/14 fields match
```

V0.1.0 · MIT · PYTHON 3.11+ · AI-ASSISTED, SEE SLIDE 10

BEFORE THE LIFT · TWO PLACES TO TRY A TRANSACTION

# THE DEV CHAIN IS EMPTY. THE TESTNET IS FINAL.

RACK · THE LIVE CHAIN

## PORTALDOT V3 TESTNET

- Real state: accounts, balances, staking, contracts.
- Every transaction is final and pays its fee in tPOT.
- The fee multiplier changed between every pair of 20 consecutive blocks (**#118,206–#118,225**).
- No public V3 node binary or chain spec was found, so V3 cannot simply be run locally.

NOTHING CROSSES

BENCH · YOUR MACHINE

## portaldot_dev --dev

- One validator, named Alice.
- None of the real chain's accounts, stakes or nominators.
- Writable, and free to break, but not the state your users are on.

A builder who wants to test one staking call, or one contract, against the state their users are actually on has no simple way to get that state onto a local node, and no way to check that what they got is faithful.

# LIFT. FORK. VERIFY. REHEARSE. REPLAY.

BAY 01

## LIFT

```
forklift up
--target v3 --at head
```

Reads every top-level storage key at one pinned block hash; **manifest_root** hashes every pair.

**5,508 keys in 4.83 s**
**2 runs, same root**

BAY 02

## FORK

```
npx @acala-network/
chopsticks@1.5.2
```

Started by **up**: V3's own runtime wasm at the pinned block, on loopback, serving the pinned genesis and **:code**.

**ready in 9.22 s**
**serves the pin: yes**

BAY 03

## VERIFY

```
forklift verify
--sample 200
```

Draws keys on the live chain and reads them on both; checks the whole key set and every lifted value.

**exit 0 · 200/200**
**5,508 = 5,508**

BAY 04

## REHEARSE

```
forklift rehearse
<Pallet.call> --send
```

Signs once, applies the bytes on the fork (the prediction), sends the same bytes to V3, compares field by field.

**3 V3 txs finalized**
**same tx hash**

BAY 05

## REPLAY

```
forklift replay
118621-1
```

Re-applies a V3 extrinsic's own bytes on a fork of its parent block and compares with the chain.

**10/10 · 14/14**
**16/16, fees equal**

CONTROL PATH

Mainnet state (spec 1002, read-only) spliced into the official **portaldot_dev** 2.0.0 node, labelled **fork-1002-fast**, not V3: fund an account, act as a real whale and nominator, and a 169-byte Fork Receipt contract.

LIVE V3 PROOF · SIGNER 5EnCf9Pa…ejvg · 2026-10-20 · ALL FINALIZED

# PREDICTED ON A FORK. THEN SENT TO V3.

### System.remark
a remark, no recipient

**118555-1**

tx 0x5b4b4d48…1a65777d

MATCHED THE PREDICTION

- ✓ same tx hash, success
- ✓ same event sequence

DIFFERED · 6 OF 10 FIELDS · EXIT 7

- fee 8,269 planck lower: in the fee, its 3 fee events and the sender's delta
- included in **#118,555**, predicted **#118,546**

REPLAY ON PARENT #118,554
**10/10** fee equal

### Balances.transfer_keep_alive
0.2 tPOT to a new account

**118621-1**

tx 0x2247c66b…73fdc36f

MATCHED THE PREDICTION

- ✓ same tx hash, success, event sequence
- ✓ recipient +0.2 tPOT, exact

DIFFERED · 6 OF 14 FIELDS · EXIT 7

- fee 3,660 planck lower: in the fee, its 3 fee events and the sender's delta
- included in **#118,621**, predicted **#118,618**

REPLAY ON PARENT #118,620
**14/14** fee equal

### Utility.batch
remark + 0.1 tPOT transfer

**118669-1**

tx 0xd371c50a…a3a3ef6a

MATCHED THE PREDICTION

- ✓ same tx hash, success, event sequence
- ✓ recipient +0.1 tPOT, exact

DIFFERED · 6 OF 16 FIELDS · EXIT 7

- fee 5,390 planck lower: in the fee, its 3 fee events and the sender's delta
- included in **#118,669**, predicted **#118,665**

REPLAY ON PARENT #118,668
**16/16** fee equal

Signed once: the bytes ran on the V3 fork first, then went to the live testnet. The fee events differ by thousands of planck; each landed 3 to 9 blocks after the predicted height, and V3's fee multiplier moves every block. **rehearse** reports it (exit 7) and never adjusts a prediction; a replay on the parent block matches field for field.

IS THE FORK FAITHFUL? CHECKED AGAINST THE CHAIN, NOT ASSUMED

# THE FORK ANSWERS TO THE CHAIN.

**V3 TESTNET · VERIFY AT #118,061 · EXIT 0**

**200/200**

draws on the live chain (198 distinct keys), all identical on the fork (seed **3145772797887710471**)

**5,508=5,508**

key set on the fork equals the lift

**5,497**

of 5,508 lifted values identical; the other 11 are child-trie roots the fork reads as null: counted and listed, not hidden

REPLAYS ON PARENT BLOCKS · FEES EQUAL TO THE PLANCK

**10/10 · 14/14 · 16/16**

The lift was run twice from fresh processes: same **manifest_root 0xfe4962de…48c430a4**. A sample is not a proof of every key, so V3 verify also compares every lifted value.

**CONTROL · FORK-1002-FAST · NOT V3**

MAINNET STATE, SPEC 1002, PIN #2,725,953

**16,324=16,324**

keys lifted = keys in an independent full listing of the same block, outside the MMR range

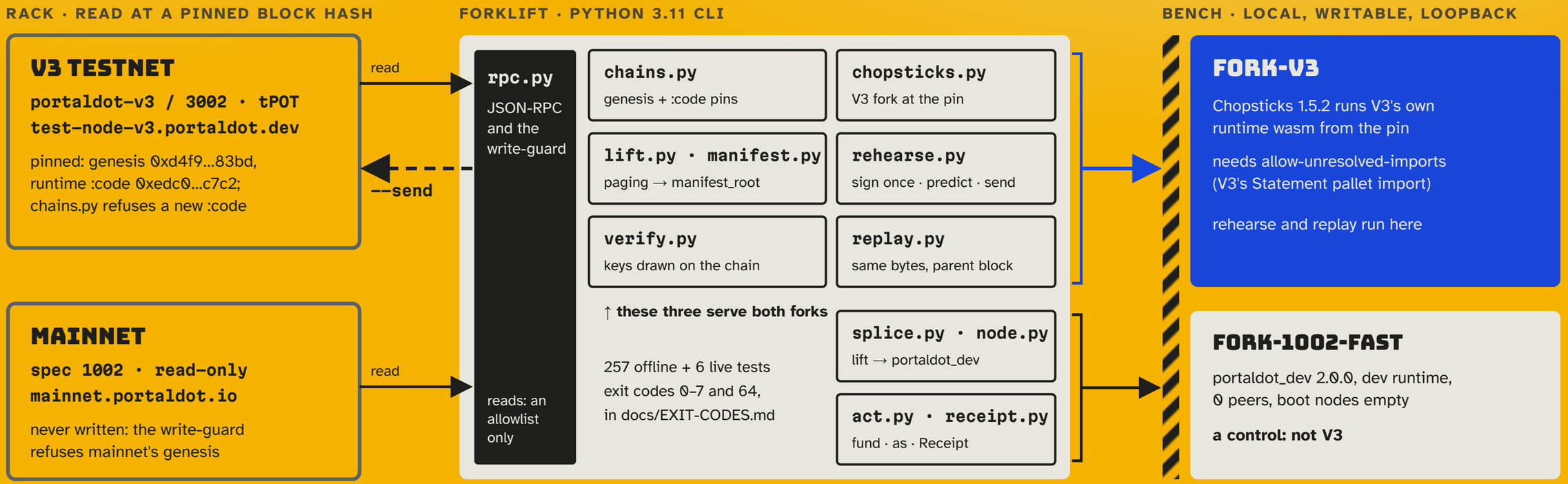
**0 missing · 0 extra**

full listing: 5,468,224 keys, 5,470 pages, 645.6 s

**exit 0**

**verify --sample 200**: 120 identical, 80 dev-kept (consensus and System keys the dev node keeps)

# RACK → FORKLIFT → BENCH




**`rehearse --send: author_submitExtrinsic only, V3 only (dashed)`**

**The one write path to a real chain. Every other write goes to a loopback fork.**

Every run writes **`evidence/*.json`**; **`scripts/proof.py`** generates **`docs/PROOF.md`** from it.

# PREDICT. SEND. COMPARE.

## ONE SIGNATURE, BOTH SIDES

The extrinsic is signed once, so the tx hash on the fork is the tx hash on V3. The prediction and the chain's record are about the same bytes.

## WRONG IS AN EXIT CODE

When the chain differs, **`rehearse`** exits 7 and lists each field. A predicted failure exits 5 and sends nothing (an unfunded signer: the fork's pool rejected it).

## FIDELITY YOU CAN RE-RUN

**`forklift replay <block>-<index>`** re-applies any V3 extrinsic on a fork of its parent block and compares it with the chain.

## PINNED BY WHAT THE CHAIN SERVES

Genesis plus runtime **`:code`** hash, not a URL. Before predicting, the fork is advanced toward the chain's best block, and the fee multiplier is re-synced if it differs.

| | V3'S REAL STATE | OUTCOME KNOWN BEFORE SENDING | CHECKED AGAINST THE CHAIN |
|---|---|---|---|
| **A dev chain** | no: Alice only | yes, on empty state | no |
| **Send to V3 directly** | yes | no: the tx is final | after the fact |
| **Forklift** | **#118,061, 5,508 keys** | **predicted, then sent: same tx hash** | **verify exit 0; replays exact** |

# BUILT ON PORTALDOT'S OWN NODES.

USES

## THE NODE, NOT A MOCK

- The historical-state RPC at a block hash: **`state_getKeysPaged`**, **`state_queryStorageAt`**.
- The official **`portaldot_dev`** 2.0.0 node, booted with zero peers.
- V3's own runtime wasm (spec 3002) on the fork.
- Both chains: the V3 testnet and mainnet state (spec 1002).

GIVES BACK · DOCS/DX-REPORT.MD

## 12

## FRICTION POINTS, EACH WITH HOW TO SEE IT AGAIN

- The macOS **`portaldot_dev`** binary is x86_64 only (runs under Rosetta).
- **`--ws-port`** is the WebSocket port, **`--rpc-port`** the HTTP port.
- Chopsticks 1.5.2 cannot fork V3 with default settings.
- V3's fee multiplier moves every block.

OPEN SOURCE

## MIT, EVIDENCE-FIRST

- 257 offline + 6 live tests.
- 83 commits to build **`44b7edc`**, dated 2026-10-14 to 2026-10-20.
- **`docs/PROOF.md`** is generated by **`scripts/proof.py`** from **`evidence/*.json`**.
- Documented exit codes and **`--json`** output, so a CI job can gate on a fork.

BUSINESS POTENTIAL

# WHO LIFTS, AND HOW IT PAYS.

### BUILDERS

Test one staking call or one contract against the state users are actually on, before spending tPOT on a transaction that is final.

### NODE OPERATORS

Rehearse an operational transaction on a fork of the chain they run, and see the events, fee and balance deltas before sending.

### WALLETS & EXCHANGES

Predict the outcome of a transaction on a pinned copy of the chain inside CI, and fail the job when the chain would differ.

**TODAY · BUILT**

A free MIT command-line tool, installed from the repository.

**NEXT · ROADMAP**

A PyPI package, a CI action and a hosted fork.

**AFTER THE EVENT · PLAN**

Paid hosted per-team forks and CI fork-as-a-service for wallets and exchanges; a shared fork, re-pinned at each runtime upgrade, offered to the Portaldot Foundation to host.

**Honest limit: no revenue and no external users yet. The business case is a funded public good plus CI minutes for integrators.**

STATUS AT THE CHECK-IN · BUILD 44b7edc

# WHAT IS DONE, WHAT IS LEFT.

## DONE AND MEASURED

- ✓ V3 profile pinned; lift, fork and verify at **#118,061**, exit 0.
- ✓ **rehearse --send**: 3 V3 transactions, all finalized.
- ✓ **replay**: all 3 equal to the chain, fees to the planck.
- ✓ Mainnet control: lift with 0 missing / 0 extra, fork, verify, fund, act as real accounts, Fork Receipt.
- ✓ 257 offline + 6 live tests · 83 commits.

## BEFORE THE FINAL

- The Fork Receipt on V3, as an ink! 5 / Revive contract.
- **--runtime exact**, a runtime-difference report, replay of a mainnet extrinsic.
- A mainnet (1002) → V3 report: which keys carry over.
- Benchmarks with p50/p95; a hosted fork, a PyPI package, a CI action.

LANDING · SCAN THE CODE
forklift.edycu.dev

REPOSITORY
github.com/edycutjong/portaldot-hackathon-2026-forklift-edycu

PROOF TABLES
docs/PROOF.md

ONE V3 TX ON NODE-CONSOLE
118621-1

**AI assistance:** Claude Code drafted most of the code, tests, README and this deck under the author's direction. The architecture, protocol choices and on-chain design were decided by the author, who reviews the code and can explain how the lift, the splice, **verify** and the Receipt work. Every figure here comes from **docs/PROOF.md**, **evidence/**, **README.md** or the git log at build **44b7edc**.